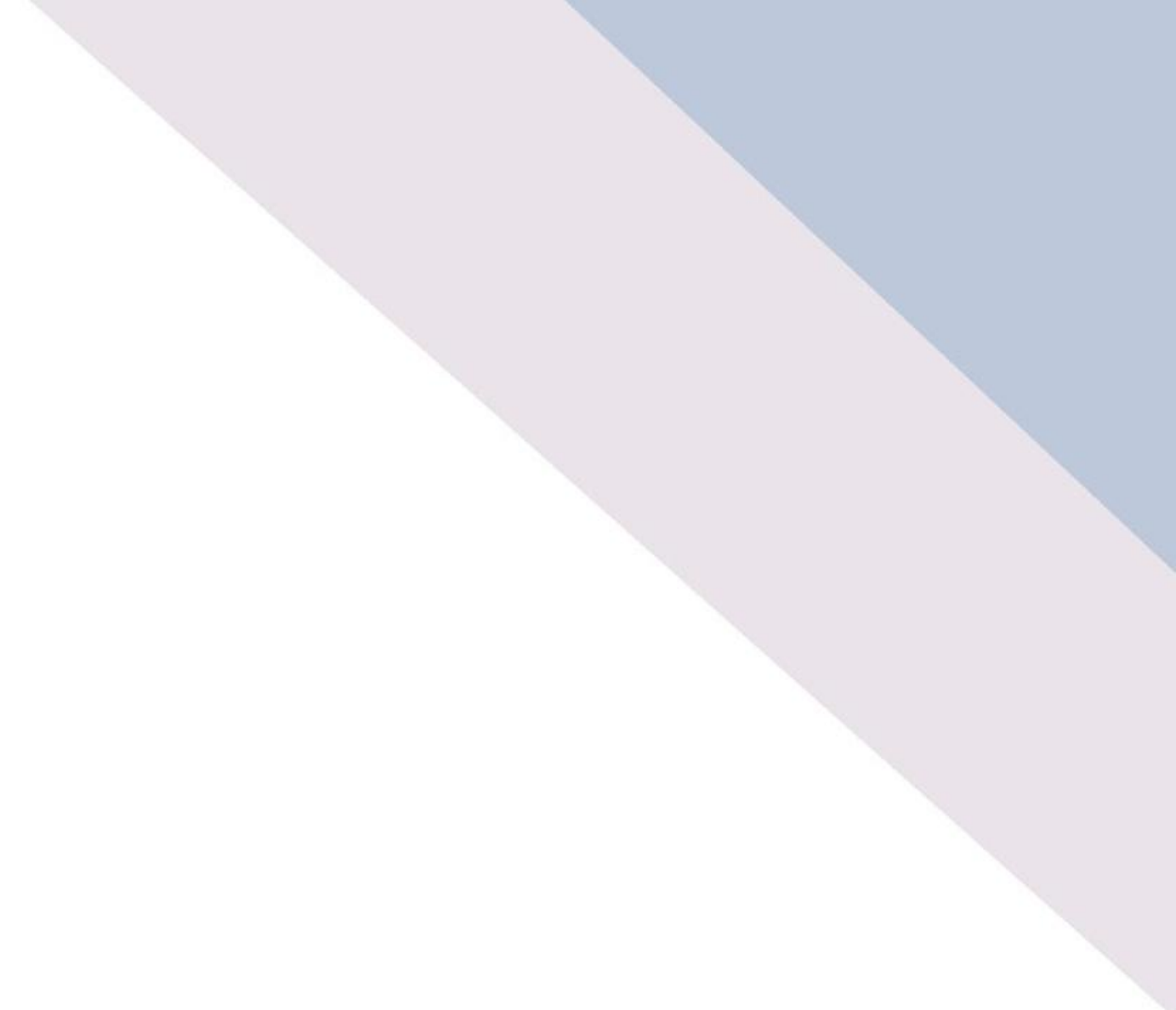


Impact of Traffic Demand on LEO NTN Performance Using SNS-3

M11502215 · Lucy

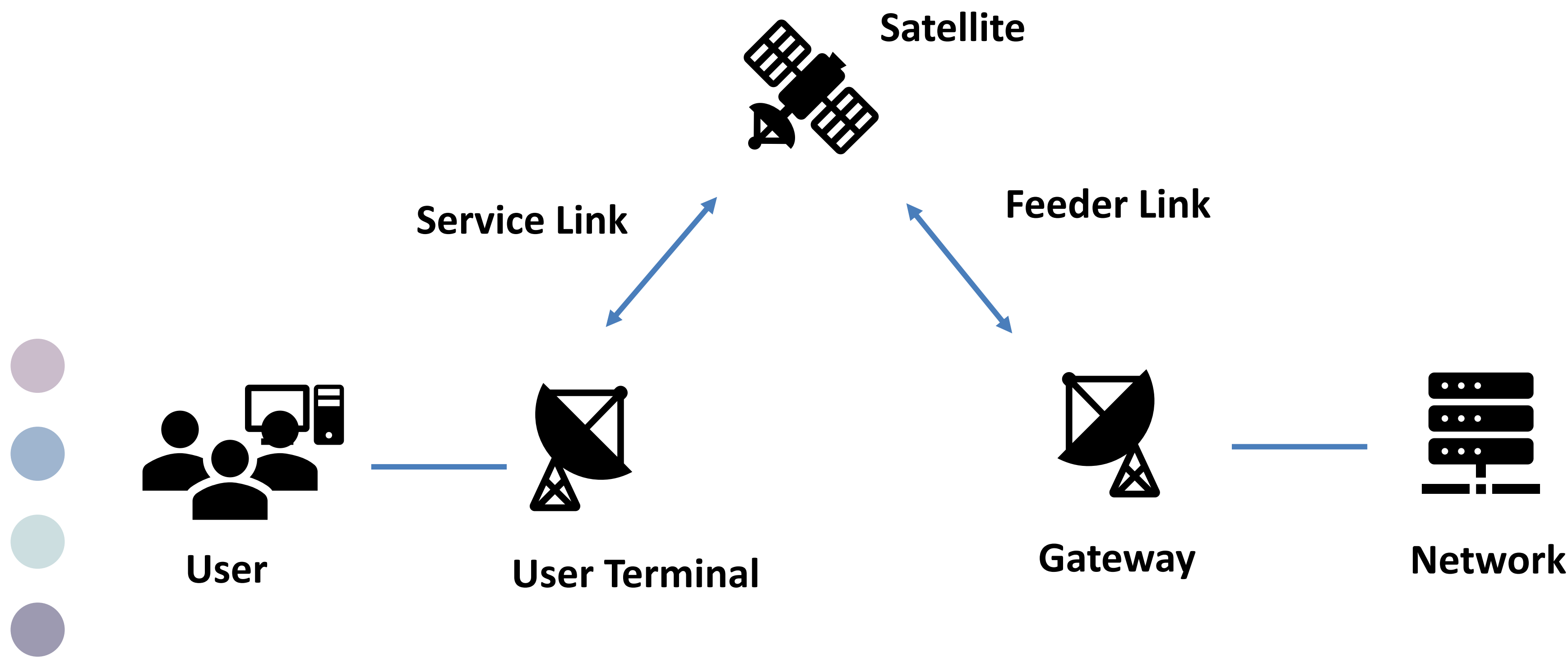
CONTENTS

01	Introduction	■
02	Reference	■
03	System Architecture	■
04	Expected Deliverables	■
		■



01 | INTRODUCTION

NTN Structure- User Plane





02

Reference

Reference

	Space/Beam Coverage	Network Architecutre	Traffic Demand	Performance Evaluation
[1]	LEO field trial	- Transparent satellites - Ground-side gNB and core network - Fixed and vehicle-mounted terminals	X	- Measured DL/UL MAC throughput - Examined NTN-specific communication effects
[2]	Defines - Earth-fixed - Quasi-Earth-fixed - Earth-moving beam coverage	- Transparent satellite - Service/feeder links.	X	- Propagation delay - Doppler shift - Link-budget
[3]	- Ground cells - Differentiated demand	Satellite network model	Models spatially and temporally differentiated traffic demand at the cell level	- Satisfaction, - Service fairness
Proposal	- Moving LEO satellites - Fixed UTs - Earth-fixed cells	- Transparent - GW → Satellite → UT	- Generate demand - Aggregate demand by cell - Vary traffic load and spatial distribution.	- Evaluates demand satisfaction - Throughput - End-to-end delay - PDR

- [1] [RECENT DEVELOPMENT ON NTN COMMUNICATION TOWARD 6G](#)
 [2] [Satellite and terrestrial Network Convergence On The Way Toward 6G](#)
 [3] [Dynamic Beam Hopping for LEO Satellites with Differentiated Traffic Demands](#)

Gap&Challenge

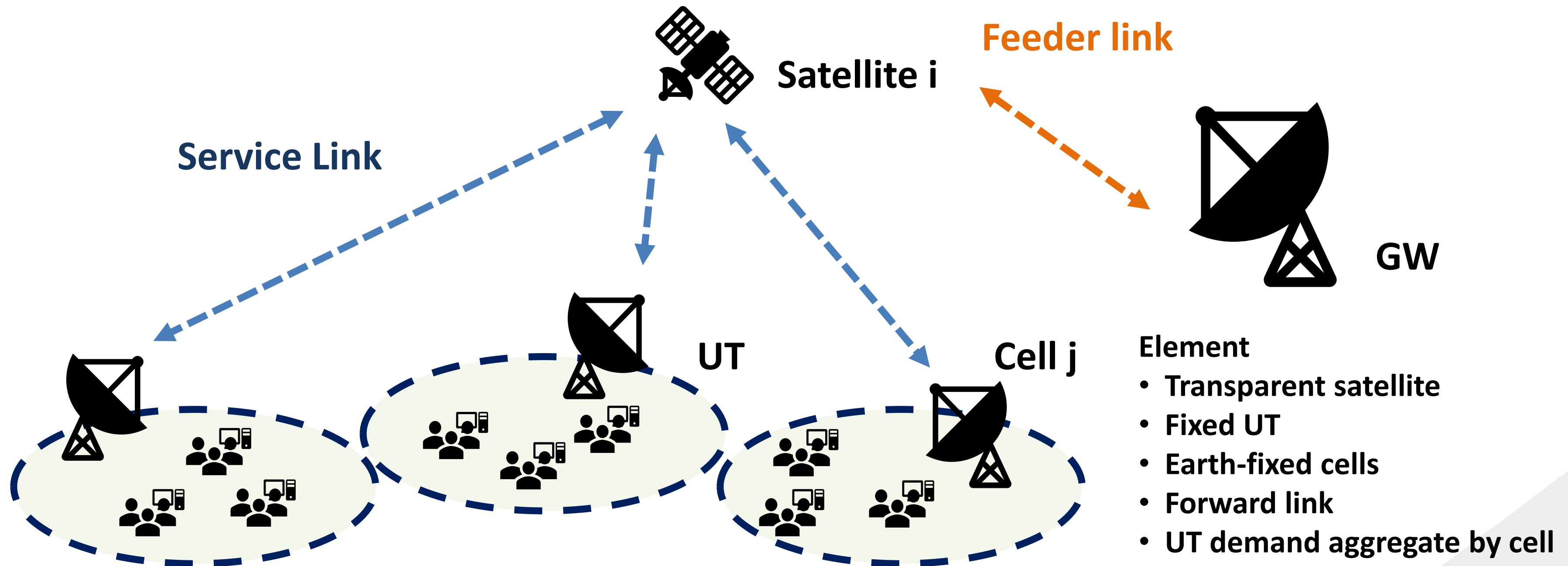
- 01** LEO mobility changes beam footprint and service coverage.
- 02** Differentiated traffic demand across geographical cells
- 03** Mismatch between offered traffic demand and delivered service



03

System Architecture

System Architecture



- i : the index of satellite
- a_j : the index of cell j

- $D_k[n]$: User k traffic demand at frame n
- $D_k^{\text{served}}[n]$: User k be served traffic at frame n

System Architecture-I/O

Input

- Topology
 - Constellation
 - User position
 - UT position
 - GW position
 - TLE
- Traffic
 - Packet size
 - Mean arrival rate (λ)



Traffic model

- Generate user demand
- User-UT mapping
- Aggregate UT demand to cell demand



- Cell demand

Output

- Demand Satisfaction Ratio
- Packet Delivery Ratio
- Throughput
- End-to-End delay



Resource allocate model

- Beam-Cell Mapping
- Allocate resources



04

Expected Deliverables

Experimental Plan & Expected Outcomes

01 Reference Scenario

Establish a transparent LEO NTN scenario with fixed UTs and a defined traffic model.

- A reproducible reference scenario for subsequent experiments.

02 Traffic Load Variation

Vary total offered traffic while keeping its spatial distribution fixed.

- Quantify how traffic load affects demand satisfaction and packet-level performance.

03 Spatial Traffic

Redistribute the same total expected offered traffic across geographical cells.

- Quantify how spatial traffic distribution affects overall and per-cell demand satisfaction and packet-level performance.

Performance Metrics

Demand Satisfaction Ratio

E2E Delay

Throughput

PDR

A decorative vertical line in a dark blue-grey color is positioned to the left of the text. In the bottom-left corner, there are two diagonal stripes: a lighter grey one on top and a darker blue-grey one below it.

Thank you